

Short Concept Questions

- 13.1. What is meant by risk-neutral valuation?
- 13.2. Explain the no-arbitrage argument used to value an option when there is a one-step binomial tree.
- 13.3. What are u and d ? What are the formulas used to calculate them?
- 13.4. How is p defined? What is the formula for calculating p ?
- 13.5. Explain the difficulty in calculating the discount rate that should be applied to an option's payoff in the real world.
- 13.6. What is the delta of a stock option?
- 13.7. What is Girsanov's theorem?
- 13.8. How is the valuation of an option using a binomial tree changed when the stock is assumed to provide a continuous dividend yield?
- 13.9. A stock price is currently \$40. It is known that at the end of 1 month it will be either \$42 or \$38. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a 1-month European call option with a strike price of \$39?
- 13.10. A stock price is currently \$50. It is known that at the end of 6 months it will be either \$45 or \$55. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a 6-month European put option with a strike price of \$50?

Practice Questions

- 13.11. A stock price is currently \$100. Over each of the next two 6-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a 1-year European call option with a strike price of \$100?
- 13.12. For the situation considered in Problem 13.11, what is the value of a 1-year European put option with a strike price of \$100? Verify that the European call and European put prices satisfy put–call parity.
- 13.13. Consider a situation where stock price movements during the life of a European option are governed by a two-step binomial tree. Explain why it is not possible to set up a position in the stock and the option that remains riskless for the whole of the life of the option.
- 13.14. A stock price is currently \$50. It is known that at the end of 2 months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a 2-month European call option with a strike price of \$49? Use no-arbitrage arguments.
- 13.15. A stock price is currently \$80. It is known that at the end of 4 months it will be either \$75 or \$85. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a 4-month European put option with a strike price of \$80? Use no-arbitrage arguments.
- 13.16. A stock price is currently \$40. It is known that at the end of 3 months it will be either \$45 or \$35. The risk-free rate of interest with quarterly compounding is 8% per annum. Calculate the value of a 3-month European put option on the stock with an exercise

price of \$40. Verify that no-arbitrage arguments and risk-neutral valuation arguments give the same answers.

- 13.17. A stock price is currently \$50. Over each of the next two 3-month periods it is expected to go up by 6% or down by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a 6-month European call option with a strike price of \$51?
- 13.18. For the situation considered in Problem 13.17, what is the value of a 6-month European put option with a strike price of \$51? Verify that the European call and European put prices satisfy put–call parity. If the put option were American, would it ever be optimal to exercise it early at any of the nodes on the tree?
- 13.19. A stock price is currently \$25. It is known that at the end of 2 months it will be either \$23 or \$27. The risk-free interest rate is 10% per annum with continuous compounding. Suppose S_T is the stock price at the end of 2 months. What is the value of a derivative that pays off S_T^2 at this time?
- 13.20. Calculate u , d , and p when a binomial tree is constructed to value an option on a foreign currency. The tree step size is 1 month, the domestic interest rate is 5% per annum, the foreign interest rate is 8% per annum, and the volatility is 12% per annum.
- 13.21. The volatility of a non-dividend-paying stock whose price is \$78, is 30%. The risk-free rate is 3% per annum (continuously compounded) for all maturities. Calculate values for u , d , and p when a 2-month time step is used. What is the value a 4-month European call option with a strike price of \$80 given by a two-step binomial tree. Suppose a trader sells 1,000 options (10 contracts). What position in the stock is necessary to hedge the trader's position at the time of the trade?
- 13.22. A stock index is currently 1,500. Its volatility is 18%. The risk-free rate is 4% per annum (continuously compounded) for all maturities and the dividend yield on the index is 2.5%. Calculate values for u , d , and p when a 6-month time step is used. What is the value a 12-month American put option with a strike price of 1,480 given by a two-step binomial tree.
- 13.23. The futures price of a commodity is \$90. Use a three-step tree to value (a) a 9-month American call option with strike price \$93 and (b) a 9-month American put option with strike price \$93. The volatility is 28% and the risk-free rate (all maturities) is 3% with continuous compounding.
- 13.24. The current price of a non-dividend-paying biotech stock is \$140 with a volatility of 25%. The risk-free rate is 4%. For a 3-month time step:
 - (a) What is the percentage up movement?
 - (b) What is the percentage down movement?
 - (c) What is the probability of an up movement in a risk-neutral world?
 - (d) What is the probability of a down movement in a risk-neutral world?
 Use a two-step tree to value a 6-month European call option and a 6-month European put option. In both cases the strike price is \$150.
- 13.25. In Problem 13.24, suppose a trader sells 10,000 European call options and the two-step tree describes the behavior of the stock. How many shares of the stock are needed to hedge the 6-month European call for the first and second 3-month period? For the second time period, consider both the case where the stock price moves up during the first period and the case where it moves down during the first period.

- 13.26. A stock price is currently \$50. It is known that at the end of 6 months it will be either \$60 or \$42. The risk-free rate of interest with continuous compounding is 12% per annum. Calculate the value of a 6-month European call option on the stock with an exercise price of \$48. Verify that no-arbitrage arguments and risk-neutral valuation arguments give the same answers.
- 13.27. A stock price is currently \$40. Over each of the next two 3-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 12% per annum with continuous compounding. (a) What is the value of a 6-month European put option with a strike price of \$42? (b) What is the value of a 6-month American put option with a strike price of \$42?
- 13.28. Using a “trial-and-error” approach, estimate how high the strike price has to be in Problem 13.27 for it to be optimal to exercise the option immediately.
- 13.29. Consider a European call option on a non-dividend-paying stock where the stock price is \$40, the strike price is \$40, the risk-free rate is 4% per annum, the volatility is 30% per annum, and the time to maturity is 6 months.
- (a) Calculate u , d , and p for a two-step tree.
 - (b) Value the option using a two-step tree.
 - (c) Verify that DerivaGem gives the same answer.
 - (d) Use DerivaGem to value the option with 5, 50, 100, and 500 time steps.
- 13.30. Repeat Problem 13.29 for an American put option on a futures contract. The strike price and the futures price are \$50, the risk-free rate is 10%, the time to maturity is 6 months, and the volatility is 40% per annum.
- 13.31. Footnote 1 of this chapter shows that the correct discount rate to use for the real-world expected payoff in the case of the call option considered in Figure 13.1 is 55.96%. Show that if the option is a put rather than a call the discount rate is -70.4% . Explain why the two real-world discount rates are so different.

Short Concept Questions

- 15.1. What does the Black–Scholes–Merton model assume about the return on a stock in a short period of time?
- 15.2. What are the mean and standard deviation of the logarithm of the stock price at time T in the model assumed by Black–Scholes–Merton?
- 15.3. In the Black–Scholes–Merton model, what happens to the standard deviation of the continuously compounded rate of return as the length of the time period over which it is measured increases?
- 15.4. How are weekends and holidays handled when volatility is estimated and used?
- 15.5. How is a riskless portfolio set up in order to derive the Black–Scholes–Merton differential equation?
- 15.6. What is the Black–Scholes–Merton price of European call and put options on an asset providing no income?
- 15.7. Explain the principle of risk-neutral valuation.
- 15.8. What is the difference between a warrant to purchase a stock and an exchange-traded stock option?
- 15.9. How is an implied volatility calculated?
- 15.10. How can the Black–Scholes–Merton formulas be adjusted to take account of known dividends?

Practice Questions

- 15.11. What does the Black–Scholes–Merton stock option pricing model assume about the probability distribution of the stock price in one year? What does it assume about the probability distribution of the continuously compounded rate of return on the stock during the year?
- 15.12. The volatility of a stock price is 30% per annum. What is the standard deviation of the percentage price change in one trading day?
- 15.13. Calculate the price of a 3-month European put option on a non-dividend-paying stock with a strike price of \$50 when the current stock price is \$50, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum.
- 15.14. What difference does it make to your calculations in Problem 15.13 if a dividend of \$1.50 is expected in 2 months?
- 15.15. A stock price is currently \$40. Assume that the expected return from the stock is 15% and that its volatility is 25%. What is the probability distribution for the rate of return (with continuous compounding) earned over a 2-year period?
- 15.16. A stock price follows geometric Brownian motion with an expected return of 16% and a volatility of 35%. The current price is \$38.
 - (a) What is the probability that a European call option on the stock with an exercise price of \$40 and a maturity date in 6 months will be exercised?
 - (b) What is the probability that a European put option on the stock with the same exercise price and maturity will be exercised?

- 15.17. Using the notation in this chapter, prove that a 95% confidence interval for S_T is between $S_0 e^{(\mu - \sigma^2/2)T - 1.96\sigma\sqrt{T}}$ and $S_0 e^{(\mu - \sigma^2/2)T + 1.96\sigma\sqrt{T}}$.
- 15.18. A portfolio manager announces that the average of the returns realized in each year of the last 10 years is 20% per annum. In what respect is this statement misleading?
- 15.19. Assume that a non-dividend-paying stock has an expected return of μ and a volatility of σ . An innovative financial institution has just announced that it will trade a security that pays off a dollar amount equal to $\ln S_T$ at time T , where S_T denotes the value of the stock price at time T .
- Use risk-neutral valuation to calculate the price of the security at time t in terms of the stock price, S , at time t . The risk-free rate is r .
 - Confirm that your price satisfies the differential equation (15.16).
- 15.20. Consider a derivative that pays off S_T^n at time T , where S_T is the stock price at that time. When the stock pays no dividends and its price follows geometric Brownian motion, it can be shown that its price at time t ($t \leq T$) has the form $h(t, T)S^n$, where S is the stock price at time t and h is a function only of t and T .
- By substituting into the Black–Scholes–Merton partial differential equation, derive an ordinary differential equation satisfied by $h(t, T)$.
 - What is the boundary condition for the differential equation for $h(t, T)$?
 - Show that $h(t, T) = e^{[0.5\sigma^2 n(n-1) + r(n-1)](T-t)}$, where r is the risk-free interest rate and σ is the stock price volatility.
- 15.21. What is the price of a European call option on a non-dividend-paying stock when the stock price is \$52, the strike price is \$50, the risk-free interest rate is 12% per annum, the volatility is 30% per annum, and the time to maturity is 3 months?
- 15.22. What is the price of a European put option on a non-dividend-paying stock when the stock price is \$69, the strike price is \$70, the risk-free interest rate is 5% per annum, the volatility is 35% per annum, and the time to maturity is 6 months?
- 15.23. Consider an American call option on a stock. The stock price is \$70, the time to maturity is 8 months, the risk-free rate of interest is 10% per annum, the exercise price is \$65, and the volatility is 32%. A dividend of \$1 is expected after 3 months and again after 6 months. Show that it can never be optimal to exercise the option on either of the two dividend dates. Use DerivaGem to calculate the price of the option.
- 15.24. A call option on a non-dividend-paying stock has a market price of \$2 $\frac{1}{2}$. The stock price is \$15, the exercise price is \$13, the time to maturity is 3 months, and the risk-free interest rate is 5% per annum. What is the implied volatility?
- 15.25. With the notation used in this chapter:
- What is $N'(x)$?
 - Show that $SN'(d_1) = Ke^{-r(T-t)}N'(d_2)$, where S is the stock price at time t and

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = \frac{\ln(S/K) + (r - \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}$$
 - Calculate $\partial d_1 / \partial S$ and $\partial d_2 / \partial S$.
 - Show that when $c = SN(d_1) - Ke^{-r(T-t)}N(d_2)$, it follows that

$$\frac{\partial c}{\partial t} = -rKe^{-r(T-t)}N(d_2) - SN'(d_1)\frac{\sigma}{2\sqrt{T-t}}$$

where c is the price of a call option on a non-dividend-paying stock.

- (e) Show that $\partial c / \partial S = N(d_1)$.
- (f) Show that c satisfies the Black–Scholes–Merton differential equation.
- (g) Show that c satisfies the boundary condition for a European call option, i.e., that $c = \max(S - K, 0)$ as $t \rightarrow T$.

- 15.26. Show that the Black–Scholes–Merton formulas for call and put options satisfy put–call parity.
- 15.27. A stock price is currently \$50 and the risk-free interest rate is 5%. Use the DerivaGem software to translate the following table of European call options on the stock into a table of implied volatilities, assuming no dividends. Are the option prices consistent with the assumptions underlying Black–Scholes–Merton?

Strike price (\$)	Maturity (months)		
	3	6	12
45	7.0	8.3	10.5
50	3.7	5.2	7.5
55	1.6	2.9	5.1

- 15.28. Explain carefully why Black’s approach to evaluating an American call option on a dividend-paying stock may give an approximate answer even when only one dividend is anticipated. Does the answer given by Black’s approach understate or overstate the true option value? Explain your answer.
- 15.29. Consider an American call option on a stock. The stock price is \$50, the time to maturity is 15 months, the risk-free rate of interest is 8% per annum, the exercise price is \$55, and the volatility is 25%. Dividends of \$1.50 are expected in 4 months and 10 months. Show that it can never be optimal to exercise the option on either of the two dividend dates. Calculate the price of the option.
- 15.30. Show that the probability that a European call option will be exercised in a risk-neutral world is, with the notation introduced in this chapter, $N(d_2)$. What is an expression for the value of a derivative that pays off \$100 if the price of a stock at time T is greater than K ?
- 15.31. Use the result in equation (15.17) to determine the value of a perpetual American put option on a non-dividend-paying stock with strike price K if it is exercised when the stock price equals H where $H < K$. Assume that the current stock price S is greater than H . What is the value of H that maximizes the option value? Deduce the value of a perpetual American put with strike price K .
- 15.32. A company has an issue of executive stock options outstanding. Should dilution be taken into account when the options are valued? Explain your answer.
- 15.33. A company’s stock price is \$50 and 10 million shares are outstanding. The company is considering giving its employees 3 million at-the-money 5-year call options. Option exercises will be handled by issuing more shares. The stock price volatility is 25%, the 5-year risk-free rate is 5%, and the company does not pay dividends. Estimate the cost to the company of the employee stock option issue.
- 15.34. If the volatility of a stock is 18% per annum, estimate the standard deviation of the percentage price change in (a) 1 day, (b) 1 week, and (c) 1 month.
- 15.35. A stock price is currently \$50. Assume that the expected return from the stock is 18% and its volatility is 30%. What is the probability distribution for the stock price in

2 years? Calculate the mean and standard deviation of the distribution. Determine the 95% confidence interval.

- 15.36. Suppose that observations on a stock price (in dollars) at the end of each of 15 consecutive weeks are as follows:

30.2, 32.0, 31.1, 30.1, 30.2, 30.3, 30.6, 33.0, 32.9, 33.0, 33.5, 33.5, 33.7, 33.5, 33.2

Estimate the stock price volatility. What is the standard error of your estimate?

- 15.37. The appendix derives the key result

$$E[\max(V - K, 0)] = E(V)N(d_1) - KN(d_2)$$

Show that

$$E[\max(K - V, 0)] = KN(-d_2) - E(V)N(-d_1)$$

and use this to derive the Black–Scholes–Merton formula for the price of a European put option on a non-dividend-paying stock.